INCLUSIVE READING ASSISTANT

HAI Principle: User Control and Agency

Empowering university students to navigate dense academic texts through explicit, user-directed AI adaptations.

The Problem

University students face severe cognitive overload when reading dense, jargon-heavy academic papers. Traditional reading takes excessive time, leading to fatigue.

The Solution

A web-based inclusive reading assistant operating strictly on a **"User Request Only"** basis, ensuring high agency before any AI intervention.

CRITICAL RULE: No adaptation happens without the user's explicit choice.

Core Features

1. Simplify

Rewrites complex paragraphs into clearer English. Aims to preserve meaning, but AI errors are possible. Original text is always restorable.

2. Summarize

Condenses long sections into core arguments. Users strictly control the length and depth of the generated output.

3. Explain

Defines highlighted jargon or methodologies contextually without altering the surrounding source text.

TARGET USER PERSONAS

Audience Definition

Amira Hassan

BELOW TYPICAL TECHNICAL LEVEL

NOVICE AI USER

ESL

Background & Context

- First-year undergraduate balancing general education.
- Reads primarily on a smartphone.
- Relies on mobile networks with sometimes slow internet.
- Hesitant about AI; worries about incorrect information.

Accessibility & Needs

- **Accessibility Need:** Benefits from reduced cognitive load, short paragraphs, and clear visual separation between the original and simplified text.
- Cognitive-load sensitivity: Easily overwhelmed.
- Requires simpler paragraph structures to parse meaning.
- Needs a seamless way to compare AI text with the original and restore it instantly.

Omar Khaled

TYPICAL TECHNICAL LEVEL

NOVICE AI USER

Background & Context

- Junior majoring in a social science discipline.
- Reads primarily on a laptop.
- Access to stable, fast internet on campus.
- Has not integrated AI tools into his daily study routine.

Accessibility & Needs

- Needs reliable summaries to triage papers efficiently.
- Requires on-demand explanations for theoretical frameworks.
- Frustration: Deeply worried about missing crucial nuances if an AI auto-summarizes.

Maya Chen

ADVANCED TECHNICAL LEVEL

EXPERIENCED AI USER

Background & Context

- Master's student conducting primary research.
- Reads on a laptop with a dual-monitor setup.
- Stable, high-speed internet.
- Understands prompting and hallucination risks deeply.

Accessibility & Needs

- Demands granular control over AI parameters (length, level).
- Requires specific technical terminology to remain intact.
- Frustration: Consumer tools strip out statistical nuance and act as a "black box".

USER JOURNEY MAPS

Interaction Flows

Amira's Journey (Simplify Focus)

1. START

Pain: High cognitive load.

Design: Clean UI. Text loads normally. AI waits for explicit input.

2. ASK

Pain: Trust & latency.

Design: Tooltip explains action. Loading indicator assures progress.

3. UNDERSTAND

Pain: Losing connection to original text.

Design: Output appears side-by-side. Source remains accessible.

4. FIX

Pain: Fear of permanent alteration.

Design: Clear "Restore Original" button ensures high reversibility.

Success: User completed task independently. Session resets cleanly.

Omar's Journey (Summarize Focus)

1. START

Pain: Wasting time on irrelevant papers.

Design: Select specific sections to summarize, not forced whole-document.

2. ASK

Pain: Lack of input context.

Design: Preference toggle before running: "Brief" vs "Detailed".

3. UNDERSTAND

Pain: Summary missing specific details.

Design: Bullet points include anchor links to exact source paragraphs.

4. FIX

Pain: AI limitation (missing facts).

Design: Anchor links let user jump to source to verify missing details.

Success: Confident triage. Export option saves summary alongside document.

Maya's Journey (Explain Focus)

1. START

Pain: Context switching to Google.

Design: User highlights specific word and clicks "Explain" inline.

2. ASK

Pain: Generic outputs unhelpful.

Design: System automatically passes surrounding paragraph as context.

3. UNDERSTAND

Pain: Tone mismatch.

Design: UI provides refinement chips: [More technical] [Give example].

4. FIX

Pain: Trapped by first-pass output.

Design: System regenerates instantly based on explicit chip feedback.

Success: Precise knowledge gained. Clicks away, resumes reading.

Measurable Project Objectives

NOTE: The following are PROPOSED TARGETS for final evaluation, not achieved results.

- **Explicit User Choice:** Proposed Target: 100% of adaptations are initiated by an explicit user choice. Zero background modifications.
- **Meaning Preservation:** Proposed Target: At least 90% of evaluated outputs preserve the original text's key meaning without hallucinations.
- **Reversibility & View Control:** Proposed Target: Users can view the original text and seamlessly restore it after adaptation in 100% of workflows.

Data Strategy Definitions

To ensure clarity in our testing and evaluation pipelines, data classifications are defined as follows:

- **Training data:** Data used to update or fine-tune a model's parameters.
- **Validation data:** Data used to select or tune a model or its configuration.
- **Test data:** Held-out data used to evaluate performance.
- **Application input:** Text entered or selected by users while using the application.

Implementation Note: This project utilizes a pre-trained LLM accessed entirely via API. **No model training or fine-tuning is performed.** Therefore, datasets like WikiLarge are not used for training; they serve strictly as *test data* for controlled experiments and evaluation of the system's logic and prompting.

Proposed Tools & Architecture

NOTE: SELECTIONS REMAIN PROPOSED AND PENDING CODEBASE IMPLEMENTATION

- **LLM (Proposed):** Existing foundational LLM accessed strictly via API calls. The model performs the requested AI tasks dynamically through the API without any local training from scratch.
- **Prompts (Proposed):** Three isolated, distinct system prompts for Simplify, Summarize, and Explain to prevent cross-contamination of instructions.
- **Security (Pending Check):** API keys will be stored securely in environment variables and never committed to Git.
- **Frontend/Backend:** Web-based React/Vue frontend routed through a Node/Python server.
- **User Control Enforcement:** The architecture mandates that the original source text remains fully accessible at all times, preserving user control over when AI operations are triggered and allowing instant reversion.
- **Source Grounding (Proposed):** The backend will pass the user-selected source text and relevant surrounding context to the LLM. Summaries and explanations should be based on the supplied text. The interface will preserve access to the original source so users can verify AI-generated claims. Source links or paragraph anchors will be included where technically supported and validated.

EVALUATION METRICS

Quality & Usability Targets

METRIC & WHAT IT MEASURES	CATEGORY	EVALUATION METHOD & TYPE	LIMITATIONS & PROPOSED TARGETS
Meaning Preservation Whether simplified output preserves the important meaning of the original.	AI QUALITY	Comparing generated output to source text key claims. *Type: Manual assessment.*	**Target:** ≥90% **Limitations:** Automatic metrics like BLEU or SARI cannot be used as the sole proof of preservation due to missing semantic nuance.
Readability Whether the output is easier to read and understand.	AI QUALITY	Reading level scores (e.g., Flesch-Kincaid) & qualitative feedback. *Type: Auto & Manual (user testing).*	**Limitations:** Readability formulas do not perfectly capture human cognitive load or contextual understanding.
Summary Quality Whether a summary captures important ideas without introducing unsupported claims.	AI QUALITY	Checking summary against original text for coverage and accuracy. *Type: Manual assessment.*	**Limitations:** Highly subjective; depends heavily on the individual reader's specific research goals.
Groundedness Whether the generated response is supported entirely by the source text.	AI QUALITY	Tracing every claim in the AI output to a supporting sentence in the source. *Type: Manual assessment.*	**Limitations:** Extremely time-consuming to evaluate comprehensively across large document sets.
Contextual Explanation Accuracy Whether explanations correctly interpret terms in their specific context.	AI QUALITY	Comparing AI explanation to domain-expert definitions as used in the passage. *Type: Manual expert review.*	**Limitations:** Requires advanced domain-expert knowledge to verify highly niche or novel jargon.
Citation / Source Alignment Whether summary statements map correctly to source passages.	AI QUALITY	Verifying generated anchor links map back to exact source citations. *Type: Manual verification.*	**Target:** ≥90% **Limitations:** Link functionality does not guarantee the semantic mapping is perfectly aligned.
Task Success Rate Whether users can successfully complete the intended reading task.	HUMAN-CENTERED	Task completion tracking (e.g., comprehension questions or triage task). *Type: User testing.*	**Target:** ≥85% **Limitations:** Laboratory testing conditions may not perfectly reflect real-world stress or fatigue.
User Control Compliance Whether AI actions occur ONLY after explicit user requests.	PRINCIPLE-LINKED	Audit ensuring original text remains accessible and zero background edits occur. *Type: Code audit.*	**Target:** 100% **Limitations:** UI dark patterns could still subvert true user agency even if technically compliant.
Error Recovery Whether users can successfully undo or	USABILITY	Tracking usage of "Restore Original" or refinement chips in simulated errors. *Type: User testing.*	**Target:** ≥90% **Limitations:** Relies on the user accurately noticing an AI error in order to initiate a recovery path.

HAI PRINCIPLES & NEXT STEPS

Framework Implementation

HAI Principle Ladder & Objectives

LEVEL	CONCRETE FEATURE	SUPPORTS AGENCY BY	MEASURABLE OBJECTIVE	TARGET & STATUS
1. INFORM	Tooltips explicitly defining Simplify vs. Summarize.	Ensuring users understand AI actions before choosing.	Users understand functions (tested via pre-task questionnaire).	**≥85% comp.** PROPOSED
2. EXPLAIN	Visual disclaimers on adapted text.	Setting expectations of AI fallibility.	Users recognize result may contain errors (post-task interview).	**≥90% recog.** PROPOSED
3. CONTROL	Manual button click. No auto-run.	Preventing unwanted background alterations.	100% adaptations initiated by explicit choice (verified by code audit).	**100% comp.** PROPOSED
4. FEEDBACK	"Restore Original" & refinement chips.	Allowing seamless recovery from unwanted results.	Users can successfully restore original text (task tracking).	**≥90% success** PROPOSED

Deliverables & Next Steps

Completed: Week 3 Plan

- Defined three distinct fictional personas.
- Mapped three end-to-end user journeys highlighting user control.
- Established measurable proposed objectives and 9 distinct evaluation metrics.
- Developed the HAI Principle Ladder v1.
- Documented Bias-Checks and Data Source proposals.

Next Milestone: Week 4

- Design the detailed conversational workflow.
- Map out comprehensive system architecture and APIs.
- Develop explicit error handling and undo paths.
- Evaluate UI against Shneiderman's Eight Golden Rules.
- **Technical Goal:** Retrieve the first real model response successfully from the backend API.

APPENDICES

Bias Checks & Data Sources

Appendix A: Bias-Check Notes

AREA REVIEWED	POTENTIAL STEREOTYPE / BIAS	CHANGE MADE / RECOMMENDED	STATUS
Names & Culture	Inferring tech ability from ethnic background.	**Change Made:** Explicitly mapped tech proficiency to experience level, not background.	REVIEWED
Accessibility & Devices	Designing only for high-end laptops, excluding low-income users.	**Recommended:** Add requirement for latency tracking and lightweight UI design (for smartphone users).	REVIEWED
Language vs. Cognition	Assuming ESL learners need "dumbed down" intellectual content.	**Recommended:** Simplify vocabulary while strictly preserving academic complexity.	RISK FLAG
User Preferences	Assuming all novice AI users want identical assistance.	**Change Made:** Included preference toggles (e.g., length control) for varying comfort levels.	CHECK

Appendix B: Data Source Description

STATUS: PROPOSED - PENDING VERIFICATION

1. Primary Dataset (Simplify Feature Evaluation)

- **Dataset Name & URL:** WikiLarge (Hugging Face Repository)
- **Dataset Size & Splits:** PENDING VERIFICATION
- **License & Permissions:** PENDING VERIFICATION (requires checking reliable source documentation for exact version)
- **Purpose & Relevance:** As detailed by Zhang and Lapata (2017), WikiLarge is purpose-built for sentence simplification. It provides aligned pairs of complex and simple sentences, representing transformations that reduce reading difficulty. It is highly suitable for conducting controlled experiments and evaluation testing for the *Simplify* feature.
- **Structure:** The dataset pairs texts formatted as 'complex' (the original sentence) and 'simple' (the simplified version).
- **Limitations:** WikiLarge consists of general sentences sourced from Wikipedia. It lacks the dense, cohesive, and prolonged academic sections necessary for testing academic summarization, nor does it provide domain-specific methodologies for testing contextual term explanation.

2. Proposed Supplementary Dataset (Summarize & Explain Features)

- **Dataset Name:** Semantic Scholar Open Research Corpus (S2ORC) *(Proposed Subset)*
- **Dataset Details:** Proposed subset size of approximately 30 open-access texts. The licenses and permitted uses of the selected papers must be verified before integration.

Appendix C: Prompt Log (Week 3)

Documenting prompts used for concept generation. Team must retrieve missing historical exact prompts.

ID	EXACT PROMPT	TOOL/ MODEL	ACTUAL OUTPUT / SUMMARY	CHANGES & REASON	STATUS
001	Exact prompt unavailable; team must retrieve from logs.	Claude	Drafted "ClearPath Reader" interaction concept.	Selected ClearPath over AdaptRead. Reason: Aligned better with User Agency.	MISSING DATA — RETRIEVAL REQUIRED
002	[Team: Insert exact prompt used for Persona generation]	[Pending]	[Team: Insert raw output summary]	[Team: Insert changes and reasoning]	MISSING DATA — RETRIEVAL REQUIRED
G001	I have an existing project document for my Human-AI Interaction (HAI) course. The document's content is mostly complete, but it has layout and formatting issues. Some content appears to overflow, extend too far down the page, or create awkward page layouts. Your task is to fix the document's layout and formatting only. Do not rewrite, summarize, remove, or change the meaning of any content. Please review the entire document page by page and: 1. Fix content that overflows page boundaries or extends awkwardly beyond the intended page layout. 2. Adjust margins, spacing, paragraph spacing, line spacing, and page breaks where necessary. 3. Resize and reposition tables, headings, text blocks, and other elements so they fit properly. 4. Keep tables readable and prevent their rows or text from being clipped or unnecessarily split across pages. 5. Maintain consistent heading styles, font sizes, alignment, indentation, and spacing throughout the document. 6. Avoid excessive blank space, awkward page breaks, isolated headings, and paragraphs that are split in visually confusing ways. 7. Preserve the existing visual identity, colors, typography, section order, page numbering, references, tables, appendices, and overall academic style as much as possible. 8. If a section genuinely needs to continue on another page, make the continuation clean and readable instead of shrinking the text excessively. 9. Check every page after making changes to ensure the final document looks professional and nothing is cut off or missing. Important restrictions: - Do not invent or add new content. - Do not delete any existing content. - Do not change technical claims, metrics, dataset information, references, or the project's proposed targets. - Do not convert pending verification items into verified facts. - Do not fabricate prompts, outputs, results, or implementation status. - Keep the document suitable for an academic HAI project submission. Return the complete corrected document in the same editable file format as the original, if supported. Also provide a PDF preview if possible. Before finishing, compare the revised document against the original to ensure that only layout and formatting have changed.	Gemini	Pending. To be recorded once the revised document is produced and reviewed.	Intended to fix layout and formatting without altering project content.	PROMPT PREPARED — OUTPUT NOT YET VERIFIED

Appendix C: Prompt Log (Week 3) (continued)

ID	EXACT PROMPT	TOOL/ MODEL	ACTUAL OUTPUT / SUMMARY	CHANGES & REASON	STATUS
C001	I am working on my Human-AI Interaction (HAI) project document, titled 'Inclusive Reading Assistant – Project Plan V1', based on the principle of User Control and Agency. Please update the existing document carefully. Your main task is to complete and organize Appendix C: Prompt Log by adding the prompts used for generating project content with Gemini and Claude, while preserving the rest of the document. Keep the existing prompt-log columns wherever possible: ID, Exact Prompt, Tool/Model, Actual Output/Summary, Changes & Reason, and Status. Add this Claude prompt as a new entry in the prompt log. Record its exact wording in the Exact Prompt column. Record Claude as the intended tool/model, but do not claim it was executed or produced an output until that actually happens. Do not invent historical prompts, outputs, model names, results, or implementation status. If the exact wording of a historical prompt used earlier to generate the personas or the initial project concept is unavailable, retain the existing missing-data status and clearly identify it as requiring retrieval. Do not present newly written prompts as historical prompts that were actually used in the past. For the new entry: - Actual Output/Summary: Pending until this task is completed and reviewed. - Changes & Reason: Organizes and documents the prompt-log update while preserving traceability. - Status: PROMPT PREPARED — OUTPUT NOT YET VERIFIED.	Claude	Pending until this task is completed and reviewed.	Organizes and documents the prompt-log update while preserving traceability.	PROMPT PREPARED — OUTPUT NOT YET VERIFIED
003	[Blank Template for Future Feature Prompts]	[Blank]	[Blank]	[Blank Template]	TEMPLATE

References and Licensing

1. **Original Research Publication:** Zhang, X., & Lapata, M. (2017). Sentence Simplification with Deep Reinforcement Learning. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing* (pp. 584-594). Association for Computational Linguistics.
2. **Dataset Repository:** WikiLarge (Hugging Face). Uploaded by *waboucay*. Available at: https://huggingface.co/datasets/waboucay/wikilarge

> **Licensing Verification Required:** The applicable license for the underlying WikiLarge dataset, Hugging Face redistribution permissions, and the S2ORC target subset require explicit confirmation. Do not proceed with full dataset integration without verifying permitted-use rights.

Requirements Compliance Checklist

Three distinct fictional personas & Journey Maps defined	COMPLETE
HAI Principle Ladder linked to testable objectives	COMPLETE
Data-source description updated (WikiLarge & S2ORC, Size, subset fit)	PENDING LICENSE CHECK
Architecture diagram & Tool selections (API vs Training verified)	PENDING CODEBASE CHECK
Exact prompt-log entries, tools, outputs	NEEDS RETRIEVAL